

DESIGNING FOR ENCOUNTER

HOW SPATIAL ANALYSIS REVEALS THE SOCIAL POTENTIAL OF CIRCULATION SPACES



TYPE: Graph Spatial Intelligence

YEAR: 2026

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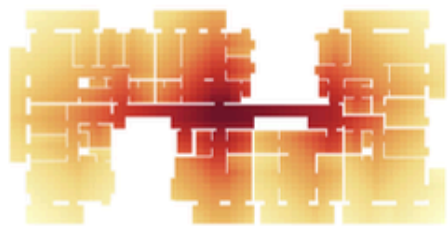
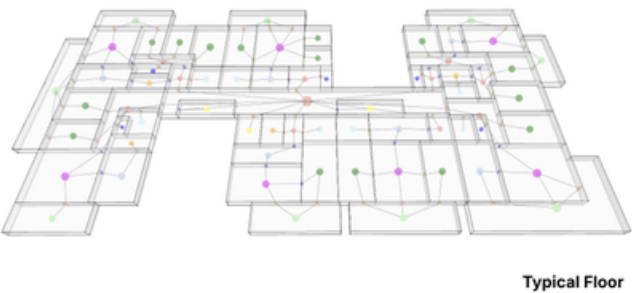
CONTEXT: IAAC Master in Advanced Computational Design (MACAD)

Traditional housing corridors are often socially inert. This paper uses spatial-network, space syntax, and visibility analyses to study three existing cohousing projects: Wohnprojekt Wien, Gleis21, and Spreefeld. The results show how circulation typologies (double-loaded, perimeter, and point-block) impact community interaction. High connectivity and visual openness systematically drive chance encounters, proving spatial analysis can shift social architecture from intuitive to data-driven design.

WOHNPROJEKT WIEN



- Room Types**
 - 1. Circulation/Entrance
 - 2. Kitchen-dining
 - 3. Living
 - 4. Bedroom
 - 5. Extra
 - 6. Bathroom
 - 7. Storage room
 - 8. Balcony
 - 9. Stairs/Elevator
- Link Types**
 - Door
 - Threshold



Typical floor



Ground floor

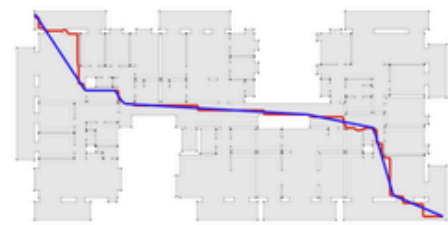
Closeness Centrality



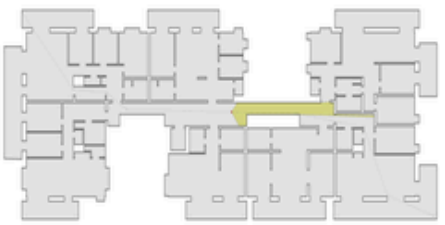
Betweenness Centrality



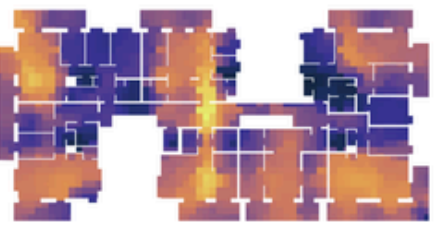
Degree Centrality



Shortest Paths



Isovists



Visibility Analysis

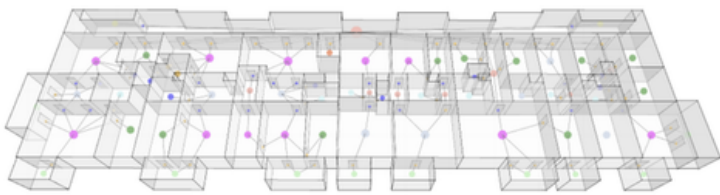
GLEIS21 WIEN



- Room Types

 - 1. Corridor
 - 2. Kitchen-dining
 - 3. Living
 - 4. Bedroom
 - 5. Bathroom
 - 6. WC
 - 7. Storage room
 - 8. Stairs
 - 9. Elevator
 - 10. Balcony
- Link Types

 - Door
 - Threshold



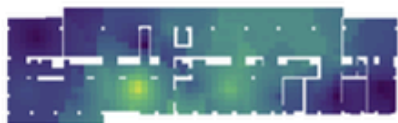
Typical Floor



Typical floor



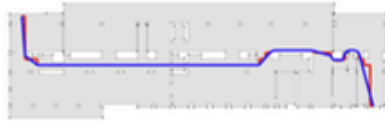
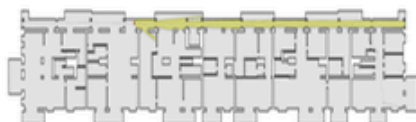
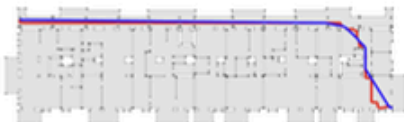
Ground floor



Closeness Centrality

Betweenness Centrality

Degree Centrality



Shortest Paths

Isovists

Visibility Analysis

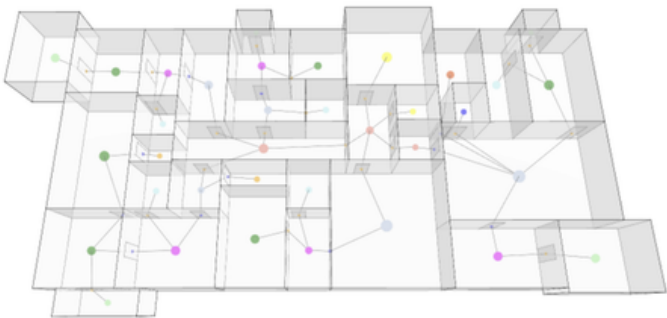
SPREEFELD BERLIN



- Room Types

 - 1. Circulation/Entrance
 - 3. Kitchen-dining
 - 4. Living
 - 5. Bedroom
 - 6. Extra
 - 7. Bathroom
 - 8. Storage room
 - 9. Balcony
 - 10. Stairs/Elevator
- Link Types

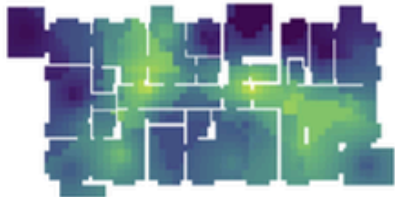
 - Door
 - Threshold



Typical Floor



Typical floor



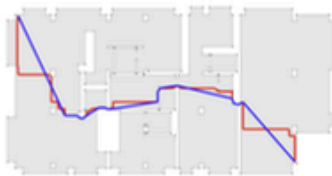
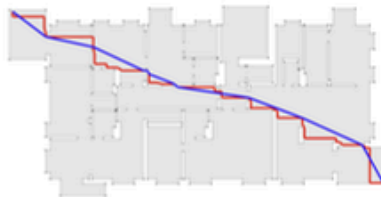
Ground floor



Closeness Centrality

Betweenness Centrality

Degree Centrality



Shortest Paths

Isovists

Visibility Analysis